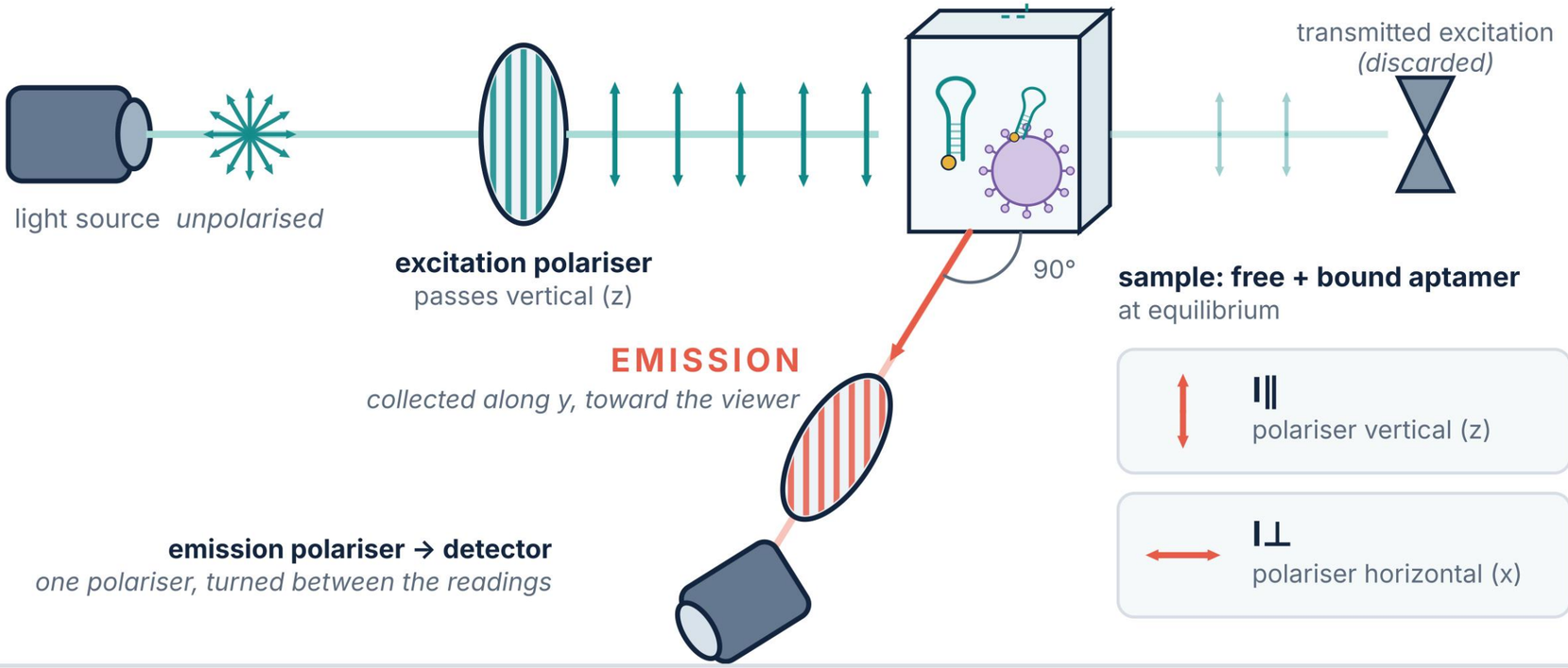
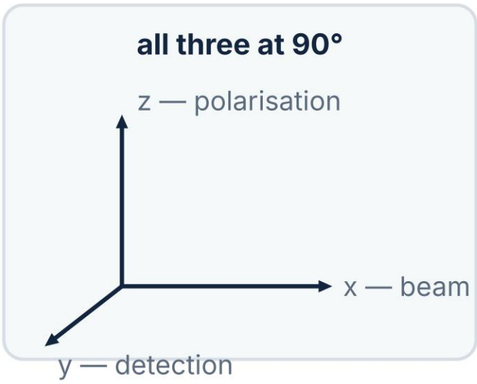
Fluorescence polarisation: how binding is read out

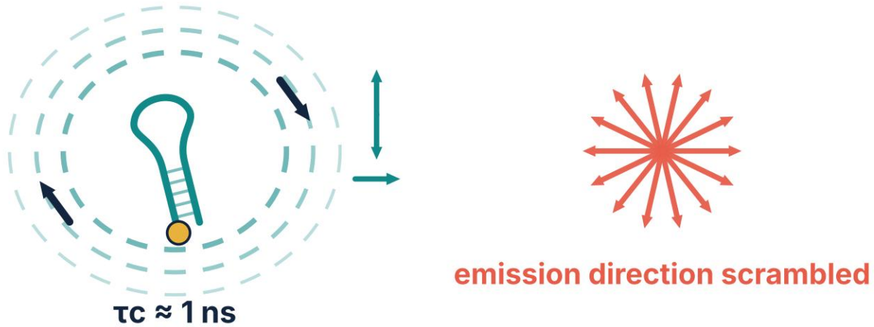
Photoselection: the vertical field excites only molecules whose dipole happens to lie along z —
that is why the emission starts out polarised





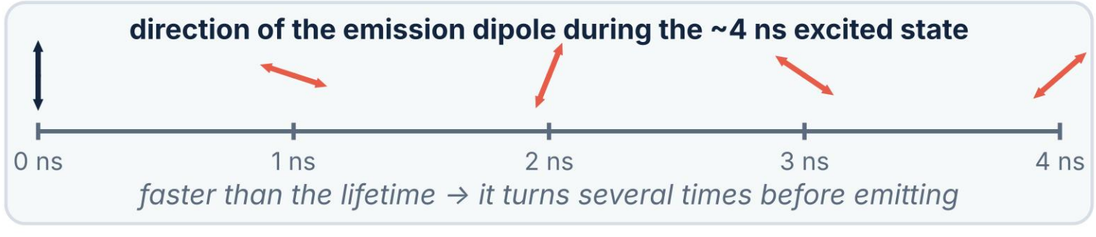
UNBOUND

free aptamer — small and quick



τc ≈ 1 ns
rotational correlation time

emission direction scrambled





LOW polarisation

P ≈ 50 mP

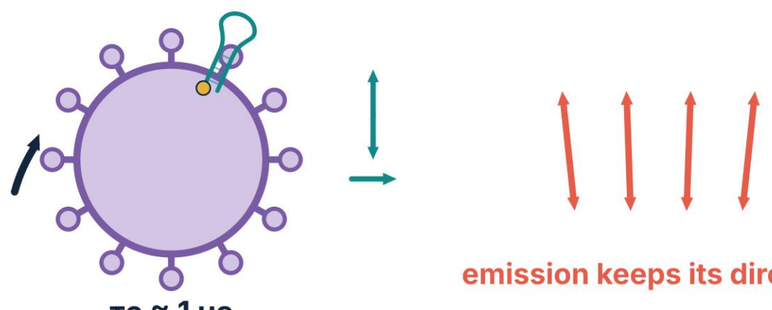
I∥ and I⊥ almost equal

$P = (I_\parallel - G \cdot I_\perp) / (I_\parallel + G \cdot I_\perp)$

$mP = 1000 \times P$

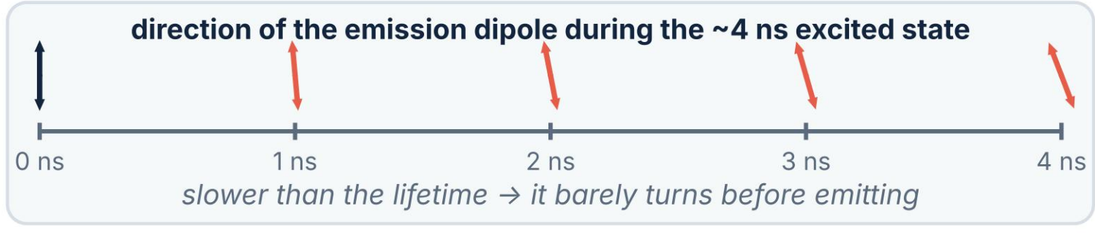
BOUND

aptamer on the virus — one big, slow object



τc ≈ 1 µs
rotational correlation time

emission keeps its direction





HIGH polarisation

P ≈ 300 mP

I∥ clearly exceeds I⊥

G corrects the detector's own polarisation bias